GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

Followup 26-14

Hypoxia refractory to supplemental oxygen

Patient	82-year-old male. Hypertension, hyperlipidemia, benign prostatic hyperplasia, chronic kidney disease. **Finished a two-week course of doxycycline for pneumonia shortly before this call** and had been feeling better. No anticoagulants, no known clotting disorder, no prior clot.
Outcome	Survived. **Large burden of bilateral pulmonary emboli**, extending from both the right and left main pulmonary arteries into the lobar and segmental branches of both lungs, with an elevated right ventricle to left ventricle ratio indicating **right heart strain** and a 1.7 cm subpleural focus in the left lower lobe favored to be a small pulmonary infarction. Troponin rose from 46 to 215 in two hours. Red critical result. Bilateral leg ultrasound found no deep vein thrombosis.
Disposition	Heparin infusion in the emergency department, intensive care admission, and catheter thrombectomy by interventional radiology the following morning, with echocardiography during the procedure showing elevated right ventricular pressures and akinesis of the right ventricular free wall with sparing of the apex. Weaned off oxygen the same morning, transitioned to apixaban, discharged home.

The call

Dispatched for shortness of breath. He was sitting up in his recliner, alert and oriented times four, **very diaphoretic**, lethargic with global weakness. His wife reported the recent pneumonia treatment and that he had finished his medication. That evening he had a **sudden onset of shortness of breath with chest pain**, and the chest pain had already resolved by the time the crew arrived.

Lung sounds were clear and equal bilaterally, with adequate rate and depth. He denied blood thinners, clots, or other significant history.

From contact	Finding or action
+0	Airway, breathing, circulation. Neurologic assessment. Head to toe. Full assessment.
+1	Oxygen 4 L by nasal cannula. 124/80, MAP 95, pulse 90, respirations 20, **SpO2 83%**.
+3	Oxygen changed to non-rebreather at 15 L. 12 lead ECG: normal sinus rhythm, no ST abnormality.
+11	148/93, MAP 111, pulse 85, respirations 20, **SpO2 82%**. Glucose 145.
+18	IV access.
+20	Cardiac monitor. Ondansetron 4 mg IV for nausea that developed in transit.
+22	Blood pressure unobtainable. Pulse 75, respirations 10, SpO2 91%.
+30	134/97, MAP 109, pulse 81, respirations 18, **SpO2 86%**.

He was no longer diaphoretic by arrival, still lethargic and weak, and unable to sign for himself.

His saturation would not come up. Room air 83 percent. Four liters, no real change. Non-rebreather at 15 L, still 82 percent eight minutes later and 86 percent at the end of transport. **Oxygen barely touched it, and his lungs were clear.**

Primary impression

Hypoxia of uncertain cause, with pulmonary embolism suspected.

The crew suspected pulmonary embolism on scene and said so at the bedside handoff. That is the right answer, reached with a stethoscope, a pulse oximeter and a history, and it deserves to be said plainly: they had the diagnosis before the hospital did.

The reasoning holds up completely. Sudden onset. Chest pain that resolved. Clear lungs with adequate ventilation. A saturation in the low eighties that did not respond to escalating oxygen. And a man two weeks into an illness that would have kept him largely off his feet. Nothing in that picture fits pneumonia, heart failure, or COPD, and the one diagnosis that explains every part of it is clot in the pulmonary circulation.

The emergency department found exactly that. The chest radiograph was effectively normal, showing only low lung volumes and bibasilar atelectasis, which is worth holding onto: **a man with clot in both main pulmonary arteries had a chest film that showed nothing.**

Hospital course

Triage found him saturating 94 percent, blood pressure 163/96, heart rate 81, afebrile at 35.8 degrees.

An arterial blood gas was drawn **on a non-rebreather at 100 percent oxygen**: pH 7.40, PCO2 46, **PO2 91**, bicarbonate 28. D-dimer 17,310. Baseline troponin 46, rising to **215 at two hours with a delta of 169**, flagged critical. Lactate normal at 1.4. White count 17.3 with neutrophilia. Creatinine 1.55 with an estimated filtration rate of 44.

CT angiography found the bilateral clot burden, the elevated right ventricle to left ventricle ratio, and the probable pulmonary infarct. Bilateral leg duplex found **no deep vein thrombosis**. Heparin was started, interventional radiology consulted, and he was admitted to intensive care. Thirty-five minutes of critical care time were documented.

The following morning he underwent catheter thrombectomy. Echocardiography during the procedure showed elevated right ventricular pressures with **akinesis of the right ventricular free wall and sparing of the apex**. He was hemodynamically stable afterward, weaned off supplemental oxygen that morning, transitioned from heparin to apixaban, and discharged home.

Incidentally, imaging six months earlier had shown a 3 cm right renal mass that was never followed up. Urology was consulted and outpatient investigation arranged, along with a hypercoagulable workup.

Final diagnosis: submassive, also called intermediate-risk, bilateral pulmonary embolism with right ventricular strain.

Education points

Why the oxygen did not work

This is the part worth understanding properly, because the failure of oxygen to correct a saturation is one of the few findings that narrows a differential to almost nothing.

Supplemental oxygen raises the concentration of oxygen in alveoli that are being ventilated. It does nothing for blood that never passes an alveolus at all. That distinction is the whole of it.

In a normal lung, ventilation and perfusion are matched: air reaches alveoli, blood flows past those same alveoli, and gas moves across. Disease breaks that match in two directions.

Dead space is ventilation without perfusion. Air reaches the alveolus, no blood arrives to collect the oxygen. That is exactly what a pulmonary embolus creates. The clot blocks the artery, the lung segment behind it keeps inflating, and it becomes useless: ventilated, unperfused, contributing nothing to gas exchange.

Shunt is perfusion without ventilation. Blood flows past alveoli that are not participating, and arrives in the systemic circulation still deoxygenated. **Oxygen cannot fix shunt**, because the shunted blood never encounters the oxygen you are giving.

A large pulmonary embolism produces both, and the second is what causes the hypoxia. When clot blocks a large fraction of the pulmonary vascular bed, the entire cardiac output has to squeeze through the vessels that remain. Those open segments are suddenly hyperperfused, with blood moving through them too fast to fully equilibrate with alveolar gas. Add the reflex redistribution of blood flow, atelectasis from loss of surfactant in the infarcted segments, and in some patients a patent foramen ovale opening under right-sided pressure, and you have blood reaching the left heart without having been oxygenated. That is functional shunt, and it is why the non-rebreather did almost nothing.

The numbers from his emergency department blood gas are stark. On a non-rebreather at 100 percent oxygen, his arterial oxygen tension was **91 mmHg**. A healthy person breathing room air runs about 95. Divide his oxygen tension by the fraction of inspired oxygen and you get a ratio of **91**, where a normal value is above 400. Working the alveolar gas equation, the alveolar oxygen tension should have been around 655, so the difference between alveolar and arterial oxygen was over 560. For an 82-year-old, the expected difference is about 25.

That gap is the diagnosis. It says the problem is not how much oxygen is going in. It is that a large fraction of his blood was bypassing gas exchange entirely.

One more feature to note. His **PCO2 was 46**, slightly high rather than low. Most patients with pulmonary embolism hyperventilate and blow their carbon dioxide down. His did not, partly because the enormous dead space made his breathing inefficient and partly because he was lethargic and, at one point in transport, breathing ten times a minute. **A normal or rising carbon dioxide in a hypoxic patient with a large embolic burden is a patient who is running out of reserve**, not a reassuring finding.

The right ventricle under acute afterload

The hypoxia is what you see. The right ventricle is what kills people.

The right ventricle is not built for pressure. It is a thin-walled chamber, roughly a third the wall thickness of the left, designed to push blood through a low-resistance circuit at about a fifth of systemic pressure. It tolerates large volumes well and sudden pressure loads badly.

Block a large share of the pulmonary arterial bed and pulmonary vascular resistance rises abruptly. The right ventricle now has to generate far more pressure than it has ever been asked for. Three things follow, and they feed each other.

It dilates. A thin-walled chamber facing a load it cannot overcome stretches rather than thickening. Dilation raises wall tension, which is what the elevated right ventricle to left ventricle ratio on his CT was measuring.

It pushes the septum into the left ventricle. The two ventricles share a wall and sit inside one fibrous sac, so they compete for space. A dilated right ventricle bows the septum leftward, and the left ventricle can no longer fill

properly. Less filling means less stroke volume, and cardiac output falls even though the left ventricle itself is perfectly healthy.

It starves itself. The right ventricle is normally perfused throughout the cardiac cycle, because pressure inside it stays low. Once it is generating high pressures, its own coronary perfusion is squeezed, at the exact moment its oxygen demand has multiplied. Falling systemic pressure from the reduced cardiac output lowers coronary driving pressure further.

That spiral is why his troponin rose from 46 to 215. He was not having a coronary event; his 12 lead showed no ST abnormality and his coronary arteries were never the problem. The troponin came from right ventricular muscle being injured by pressure and ischemia. It is an index of strain, and in pulmonary embolism it is a prognostic marker rather than a diagnostic one.

The echocardiogram during his thrombectomy showed the classic pattern: akinesis of the right ventricular free wall with the apex still moving. That specific combination, free wall failure with apical sparing, is known as McConnell's sign and is fairly characteristic of acute right ventricular pressure overload.

One practical consequence worth carrying. A failing right ventricle is exquisitely sensitive to further loading. **Aggressive fluid in a patient with right heart strain can make things worse**, by over-distending a chamber that is already stretched and pushing the septum further into the left ventricle.

Risk stratification, and why he went to intensive care

Pulmonary embolism is sorted into three bands, and the difference between them is not clot size.

Low risk: normal blood pressure, no right ventricular strain, normal cardiac biomarkers. Anticoagulation, often as an outpatient.

Intermediate risk, also called submassive: normal blood pressure, but evidence of right ventricular strain on imaging or echocardiography, or raised troponin or natriuretic peptide. He had strain on CT and a critical troponin with a normal blood pressure, which is exactly this band. Mortality here is substantially higher than in the low-risk group, and these are the patients considered for catheter-directed treatment.

High risk, also called massive: sustained hypotension or shock. These patients get thrombolysis or mechanical intervention urgently.

His blood pressures were normal all night, 124/80 to 148/93 in the field and 163/96 at triage. Nothing about his pressure suggested a dying patient. What put him in intensive care was the combination of strain on imaging and a troponin that had more than quadrupled in two hours.

Two of his numbers were normal and should not have been reassuring. His **lactate was 1.4**, meaning he was still perfusing tissue adequately, which is what compensation looks like before it fails. And his **pro-BNP was 44**, entirely normal, in a man with documented right heart strain. Natriuretic peptide is released in response to sustained myocardial stretch and takes hours to rise; his event had started that evening. **A normal natriuretic peptide early in an acute presentation does not exclude ventricular strain.**

Bilateral leg ultrasound found no deep vein thrombosis. That does not argue against the diagnosis at all. Roughly half of patients with confirmed pulmonary embolism have no detectable leg clot, for the straightforward reason that the clot has already left the leg and is now in the lung.

> **Further reading on right ventricular failure in pulmonary embolism.** The mechanism, the imaging findings and the management decisions go further than a case document can carry. This open-access review covers it well: https://www.ccjm.org/content/92/5/301

Prehospital considerations

This crew recognized the pattern and acted on it, and they had it right. They told the receiving physician they suspected pulmonary embolism, and they arrived at that from three observations that all pointed the same way: sudden onset, clear lungs with adequate ventilation, and a saturation that would not respond to oxygen. They had recently been through the ventilator training where refractory hypoxemia and shunt physiology are worked through in detail, and it shows in the record. **That is training turning into a bedside diagnosis on the next call, which is exactly what it is for.**

Hypoxia with clear lungs narrows the list fast. Most causes of low saturation make noise: crackles, wheeze, absent sounds on one side. When the chest is clear and the patient is moving air well and the number is still in the eighties, the problem is not in the airways or the alveoli. It is in the blood flow or the blood itself. That list is short: pulmonary embolism, shunt, anemia, methemoglobinemia, or carbon monoxide.

Oxygen that does not work is diagnostic information. Escalating from nasal cannula to non-rebreather and getting nothing is not a treatment failure to be embarrassed about. It is a positive test result, and it belongs in the handoff in those words.

Capnography would have added to this. A large embolus creates dead space, so exhaled carbon dioxide falls relative to arterial carbon dioxide. A low end-tidal reading in a patient who is breathing adequately with clear lungs supports the same diagnosis the saturation is pointing at, and the gap between the two widens as clot burden rises.

Recent illness and immobility count as risk factors. He denied clots and blood thinners and had no formal history, so on a checklist he scored low. But two weeks of pneumonia in an 82-year-old means two weeks of sitting still, and that is a risk factor even when nothing in the past medical history is.

His heart rate never rose. 90, 85, 75, 81, across thirty minutes of saturations in the eighties. Tachycardia is the textbook sign and he never had it. An older patient may not produce one, and its absence is not evidence against the diagnosis.

The interventions

PREHOSPITAL INTERVENTIONS

Oxygen, escalated from nasal cannula to non-rebreather. Correct, and the failure to respond is the finding. Worth noting the narrative records the non-rebreather at 2 L while the events list records 15 L. A non-rebreather will not fill its reservoir below about 10 to 15 L, and the events list is the record.

12 lead ECG. Normal sinus rhythm with no ST abnormality, which took acute coronary syndrome off the table early in a patient with chest pain, diaphoresis and a rising troponin still to come.

Ondansetron 4 mg IV for nausea that developed in transit, with improvement documented.

Serial vital signs, four sets in thirty minutes. The trend is what makes the case legible, and the saturation trend in particular is what demonstrates refractory hypoxemia rather than a single low reading.

The verbal handoff. The crew's statement that they suspected pulmonary embolism is the single most valuable intervention on this call. CT angiography was ordered, the clot was found, and heparin was running within the hour.

IN-HOSPITAL INTERVENTIONS

Arterial blood gas on 100 percent oxygen. The test that quantified the shunt, and the reason the numbers above can be worked at all.

CT angiography of the chest. Contrast timed to the pulmonary arteries, which shows clot directly and simultaneously measures the right ventricle to left ventricle ratio. One scan gave both the diagnosis and the risk stratification.

Heparin infusion. Unfractionated heparin accelerates antithrombin, which inactivates thrombin and factor Xa. It does not dissolve existing clot; it stops propagation while the body's own fibrinolysis works. Unfractionated heparin was chosen over a low molecular weight agent because it can be switched off quickly, which matters when a procedure is planned and his kidney function was reduced.

Catheter thrombectomy. A catheter advanced through the venous system into the pulmonary arteries to remove clot mechanically. It is offered in intermediate-risk embolism to unload the right ventricle without the bleeding risk of systemic thrombolysis, which matters in an 82-year-old.

Transition to apixaban, a direct factor Xa inhibitor, for ongoing oral anticoagulation.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Lab	Value	Reference range
PO2, arterial, on 100% oxygen (mmHg)	91	83 to 108
PCO2, arterial (mmHg)	46	35 to 45
pH, arterial	7.40	7.35 to 7.45
D-dimer (ng/mL FEU)	17,310	0 to 500
hs troponin T, baseline (ng/L)	46	22 or less
hs troponin T, 2 hour (ng/L)	215	22 or less
Delta from baseline (ng/L)	169	Less than 8
Pro-BNP (ng/L)	44	125 or less
Lactic acid (mmol/L)	1.4	0.0 to 2.0
White blood cell (x10^9/L)	17.3	4.0 to 10.8
Neutrophils, absolute (x10^9/L)	11.68	1.80 to 7.80
Creatinine (mg/dL)	1.55	0.70 to 1.20
Hematocrit, point of care (%)	52.0	38.0 to 50.0

The oxygen tension of 91 on 100 percent oxygen is the most important number here, and it is the one that looks most normal. Taken alone it is a perfectly acceptable value. Taken against the oxygen he was receiving, it describes a lung through which most of the blood was passing without picking up oxygen.

The D-dimer at 17,310 is thirty-five times the cutoff. D-dimer is a fragment released when cross-linked fibrin is broken down, so it measures clot formation and breakdown anywhere in the body. Its value is in ruling out rather than ruling in, because it rises in infection, malignancy, surgery, pregnancy and old age. A number this large in this context is not subtle.

The troponin quadrupled in two hours. 46 rising to 215 with a delta of 169, flagged critical. Right ventricular muscle injured by pressure, not a coronary event.

The white count of 17.3 with neutrophilia is not infection. His procalcitonin was 0.13 and his blood cultures grew nothing. Acute pulmonary embolism produces an inflammatory and stress response, and infarcted lung contributes. He had also just completed treatment for pneumonia, which makes this the exact number that could send someone back down the infection path.

Hematocrit of 52 is hemoconcentration, consistent with an unwell older man with reduced intake over two weeks.

Learning points

1. **Hypoxia that oxygen will not correct, in a patient with clear lungs, is shunt until proven otherwise.** Oxygen raises the concentration in ventilated alveoli and does nothing for blood that bypasses them. This patient went from 83 percent on room air to 82 percent on a non-rebreather, and his arterial oxygen tension on 100 percent oxygen was 91 when it should have been over 600.
2. **The right ventricle is what kills in pulmonary embolism.** A thin-walled chamber facing a sudden pressure load dilates, pushes the septum into the left ventricle, reduces cardiac output, and then starves its own coronary supply. His troponin rose from 46 to 215 out of right ventricular muscle, with a completely normal 12 lead.
3. **Normal biomarkers early do not exclude strain.** His pro-BNP was 44 and his lactate was 1.4, both normal, in a man with documented right heart strain and a critical troponin. Natriuretic peptide takes hours to rise, and a normal lactate means compensation is still working.
4. **A negative leg ultrasound does not argue against pulmonary embolism.** About half of confirmed cases have no detectable leg clot, because it has already embolized. His duplex was clean in both legs.
5. **Recent illness is a risk factor even when the history is empty.** He denied clots, blood thinners and clotting disorders, and he had spent two weeks largely immobile recovering from pneumonia. On a scoring tool he looked low risk. On the physiology he was not.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at https://www.berrienmca.org/followup/